Drift-VLA: Direct One-Step VLA Action Generation via Action-Dimension Drifting

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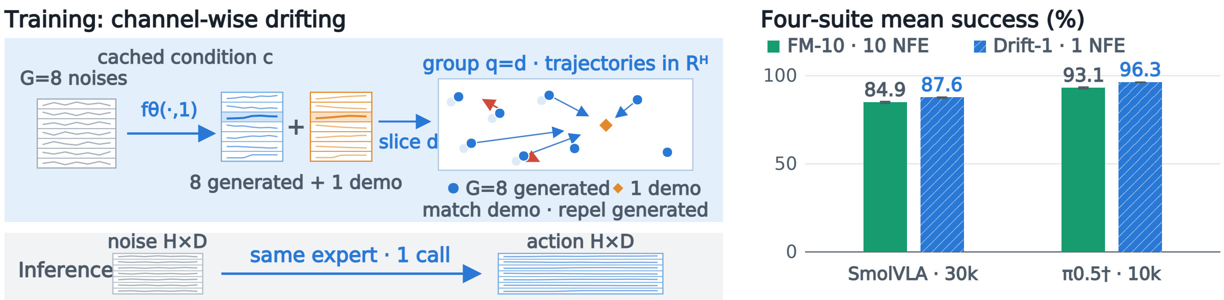

Figure 1. Left: training slices G=8 generated chunks and the demo into channel-wise $\mathbb{R}^H$ metric groups—not separate experts. The stopped target attracts to the demo and repels generated samples with local normalization; inference uses one call without G expansion. Right: descriptive unweighted four-suite means from one training run/objective; compare within family. Bars show means; whiskers are sample SD over three evaluation-seed suite means. $^\dagger\pi_{0.5}$ is not padding-loss-matched; protocols are in Table 1.

Abstract

Flow-matching (FM) vision–language–action (VLA) policies repeatedly evaluate an action expert per chunk. DRIFT-VLA maps noise directly to a chunk with the same action-head architecture, without a teacher, second training stage, or added parameters. For padded $H \times D_{\max}$ chunks, it gives each physical channel's H-step trajectory its own matching group. In a VLM-frozen SmolVLA study on the LIBERO-Spatial development suite, action-dimension grouping reaches 90.2/91.2% success at 30k across two matched training seeds, versus 77.8/82.0% for flat-chunk grouping. Separately, three matched training seeds under full fine-tuning, each evaluated on Spatial at one fixed evaluation seed, yield 94.2% for Drift-1 versus 94.3% for FM-10; the paired interval is too wide to establish equivalence. Grouping varies only in the VLM-frozen Spatial study; four-suite checks use one training run per objective in both model families, though the $\pi_{0.5}$ pair is padding-loss-confounded. Batch-one action-generation latency falls from 245.6 to 132.4 ms on an RTX 2050 (1.86×); separate A100 protocols show 3.0–4.4×. These descriptive results support recipe feasibility and inference savings, not objective superiority.

1. Introduction

Diffusion and flow matching provide expressive stochastic robot action heads (Lipman et al., 2023; Chi et al., 2023). In both SmolVLA and $\pi_{0.5}$, roughly ten expert evaluations per chunk follow a single vision–language model (VLM) prefix (Shukor et al., 2025; Black et al., 2025a). Prefix cost limits the wall-clock gain from fewer network function evaluations (NFEs), but decoding remains material. Execution-time methods can hide delay (Black et al., 2025b); one-step generation removes the repeated calls.

We instantiate the explicit finite-G Drift-Based Policy (DBP) estimator (Gao et al., 2026) for padded, heterogeneous VLA chunks. Its attraction toward the demonstration and repulsion among generated samples depend on a metric that determines which coordinates share a geometry; DBP evaluates one metric over the full chunk or one per timestep. SmolVLA exposes a padded $H \times D_{\max} = 50 \times 32$ chunk for $D = 7$ semantically heterogeneous physical channels (translation, rotation, and gripper). DRIFT-VLA slices unused coordinates, masks temporal padding, and assigns each channel's full H-step trajectory its own matching problem. This retains within-channel cross-time structure without forcing one distance scale across channels.

Contributions. (1) A D-block finite-G DBP factorization with per-channel distance/force normalization. (2) A two-training-seed, two-checkpoint flat-versus-action-dimension study on VLM-frozen Spatial. (3) Three-training-seed Spatial replication, single-run four-suite checks in two models, and 1.86–4.4× action-generation speedups.

2. Related Work

One-step generators use distillation (Prasad et al., 2024; Wang et al., 2025; Liu et al., 2026a), teacher-free/self-distilled objectives (Su et al., 2025; Chen et al., 2026a; Zhan et al., 2026; Chen et al., 2026c; Li et al., 2026; Luan et al., 2026; Chen et al., 2026d), or regression/token heads (Kim et al., 2025; Liang et al., 2026). NinA and an invertible adapter decode flows once; FASTER finalizes near-term actions in one step while refining later ones (Tarasov et al., 2025; Zhang et al., 2026; Lu et al., 2026). FreqPolicy aligns discrete-cosine horizon features across flow times; High-Fidelity margin-separates with in-batch negative target velocities; FASTerVQ groups physical types into time–dimension tokens (Su et al., 2025; Chen et al., 2026b; Liu et al., 2026b). DBP uses whole chunks or timestep slices; Ada3Drift flattens trajectories (Gao et al., 2026; Xu et al., 2026). Ours factorizes DBP's estimator into one H-step group per channel. Keyed Drifting Policies (KDP) builds key-space neighborhoods over full trajectories; Implicit Drifting Policy (IDP) learns diagonal precision from local/global expert geometry (Puthumanaillam & Ornik, 2026; Yang et al., 2026). IDP's deterministic $G=1$ analysis yields isotropic MSE for positive translation-invariant kernels; ours couples H coordinates per block at $G=8$. Other work reformulates the drift field (Koo et al., 2026). We claim neither one-step training nor a new general drifting objective.

3. Action-Dimension Drifting

Flow-matching baseline. For a demonstration chunk a, conditioning c, and Gaussian noise ϵ, the baseline samples $u \sim \mathrm{Beta}(1.5, 1)$, sets $t = 0.999u + 0.001$, and trains a velocity field along $x_t = t\epsilon + (1-t)a$,

$$\mathcal{L}_{\text{FM}} = \frac{1}{|\Omega|} \sum_{(h,d)\in\Omega} \left[v_\theta(x_t, t, c) - (\epsilon - a)\right]^2_{h,d}. \tag{1}$$

Here Ω is the set of sliced, temporally valid coordinates for SmolVLA; the archived $\pi_{0.5}$ FM path instead averages all $H\times D$ entries. Inference integrates backward from $t=1$ noise to $t=0$ actions with ten Euler evaluations.

Partition into matching groups. For $g \in \mathbb{R}^{H\times D}$, let (Q, S_q) denote the number of groups and size of group q. Timestep, flat-chunk, and action-dimension grouping use $(Q, S_q) = (H, D), (1, HD), (D, H)$, respectively. DBP studies the first two; ours normalizes distance and force per channel trajectory. Partitioning jointly changes (Q, S_q) and groupwise normalization; generation, kernel, temperatures, and optimizer remain fixed.

Implemented drifting objective. Following DBP's conditional construction (Gao et al., 2026), for minibatch observation b with conditioning c_b, we cache one VLM prefix and expand it to generate $G = 8$ chunks $g_{bi} = f_\theta(\epsilon_{bi}, t{=}1, c_b)$, $i = 1,\ldots,G$. For group q, let g_{qbi} and a_{qb} be the corresponding S_q-dimensional generated and demonstration slices. Set $\tilde{g} = \mathrm{sg}(g)$, where sg stops gradients, and $\mathcal{Y}_{qb} = (\tilde{g}_{qb1}, \ldots, \tilde{g}_{qbG}, a_{qb})$. Let $d_{qbij} = \sqrt{\max\{\|\tilde{g}_{qbi} - \mathcal{Y}_{qbj}\|_2^2, 10^{-8}\}}$ be the generated-to-candidate distances and $\bar{d}_q$ their mean over all pairs and valid observations in a device-local minibatch (no cross-rank synchronization). Define $\delta_q = \max(\bar{d}_q, 10^{-3})$ and $\sigma_q = \max(\bar{d}_q/\sqrt{S_q}, 10^{-3})$, and $\tilde{d}_{qbij} = d_{qbij}/\delta_q + 100\,\mathbf{1}[j = i \le G]$, where $\mathbf{1}[\cdot]$ is the indicator function. For temperature R, write $r^R = \mathrm{softmax}_{\text{row}}(-\tilde{d}/R)$ and $s^R = \mathrm{softmax}_{\text{col}}(-\tilde{d}/R)$, with softmax over candidate index j and generated-sample index i, respectively. Let $\mathcal{G} = \{1,\ldots,G\}$ and $\mathcal{P} = \{G+1\}$ index generated-sample and demonstration-positive candidates. With ℓ indexing group coordinates, the implemented affinity and mass-coupled force (Deng et al., 2026; Gao et al., 2026) are

$$\begin{aligned}
A^R_{qbij} &= \sqrt{\max\left\{r^R_{qbij} s^R_{qbij}, 10^{-6}\right\}},\\
M^{R,X}_{qbi} &= \sum_{k\in X} A^R_{qbik}, \quad X \in \{\mathcal{G}, \mathcal{P}\},\\
w^R_{qbij} &= A^R_{qbij}\left(M^{R,\mathcal{G}}_{qbi}\mathbf{1}[j\in\mathcal{P}] - M^{R,\mathcal{P}}_{qbi}\mathbf{1}[j\in\mathcal{G}]\right),\\
F^R_{qbi} &= \sum_{j=1}^{G+1} w^R_{qbij}(\mathcal{Y}_{qbj} - \tilde{g}_{qbi})/\sigma_q,\\
\rho^R_q &= \sqrt{\max\left\{\operatorname*{mean}_{(b,i,\ell)\text{ valid}}[(F^R_{qbi,\ell})^2], 10^{-8}\right\}},\\
V_{qbi} &= \sum_{R\in\{.02,.05,.2\}} F^R_{qbi}/\rho^R_q.
\end{aligned} \tag{2}$$

The affinity A^R is not doubly stochastic; $+100$ softly penalizes self-edges; ρ^R_q and $\bar{d}_q$ are rank-local. The quantities in Equation 2 and the target below are detached. For each valid group–observation pair, the loss is

$$\mathcal{L}_{qb} = \frac{1}{GS_q}\sum_{i=1}^{G}\left\|\frac{g_{qbi}}{\sigma_q} - \mathrm{sg}\left(\frac{g_{qbi}}{\sigma_q} + V_{qbi}\right)\right\|_2^2, \tag{3}$$

The implementation averages $\mathcal{L}_{qb}$ over valid (q,b); its gradient is proportional to $-V_{qbi}/\sigma_q$. Generated and demonstrated actions are upcast to at least fp32; targets are formed outside autocast. Invalid timestep (q,b) pairs are excluded from statistics and loss; flat/action-dimension layouts zero-mask padding with fixed S_q. Chunks are first sliced to D (7 of 32 on LIBERO). At $t=1$, the expert output is treated as the normalized chunk, then denormalized—no Euler or residual update. The method adds no parameters; training uses $G=8$ suffixes per condition and an $O(G^2)$ matcher. Without training-cost controls, efficiency claims are inference-only.

4. Experiments

Protocol overview. All success results use the official LIBERO evaluator; comparisons are within model family and fixed recipes. Table 1 reports suite breadth; Table 2 reports grouping behavior; Fig. 2(a) compares $\pi_{0.5}$ training settings; Fig. 2(b) reports latency under separate protocols.

Table 1. Fixed-checkpoint success (%) on four LIBERO suites and separate A100 action-generation latency. Checkpoints follow full fine-tuning on all 40 tasks, not unseen-suite transfer. Success is mean ± *sample* SD over three evaluation seeds from one training run per row; Avg. is the within-seed suite mean. Episodes/suite/seed: 500 for SmolVLA; 200 (Spatial) or 500 (others) for $\pi_{0.5}$. Each replan executes one action step for SmolVLA and ten for $\pi_{0.5}$. Latencies are batch-one architecture microbenchmarks, not end-to-end measurements. *: raw SmolVLA timing records unavailable. †: $\pi_{0.5}$ is not padding-loss-matched (Drift masks padding; FM averages $H \times D$).

Model	Objective	NFE	Spatial	Object	Goal	Long	Avg.	ms/chunk
SmolVLA	FM	10	94.1±0.2	83.9±1.3	87.5±0.8	74.0±1.6	84.9±0.5	222.6*
	Drift	**1**	94.4±1.2	92.5±0.8	89.2±1.0	74.3±0.8	87.6±0.3	50.6*
$\pi_{0.5}^{\dagger}$	FM	10	96.2±2.0	96.8±0.4	92.6±0.3	86.7±1.4	93.1±0.5	259.8
	Drift	**1**	98.3±0.3	98.1±0.9	95.6±0.8	93.1±0.6	96.3±0.1	**85.6**

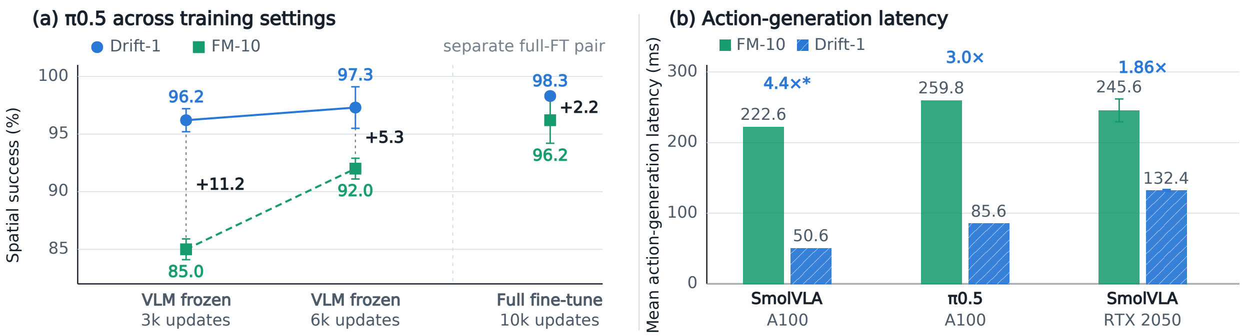

Figure 2. Training settings and representative action-generation latency. **(a)** $\pi_{0.5}$ Spatial. Lines join 3k/6k checkpoints of the same two VLM-frozen runs; full-FT is separate. Points are mean ± sample SD over three evaluation seeds, with one training run per objective/regime. Recipes differ in updates/modules/batch; frozen FM still improves and the arms are not padding-loss-matched. Gaps use unrounded means and are descriptive, not capacity/asymptotic effects. **(b)** Batch-one latency: A100 values are means; RTX 2050 error bars show reported SD over five round medians. Protocols are not hardware-matched; raw RTX and * SmolVLA records are unavailable.

Table 2. VLM-frozen SmolVLA Spatial grouping ablation; partitioning changes (Q, S_q) and groupwise normalization (generation, kernel, temperatures, and optimizer fixed). Values are success (%): mean ± sample SD over three evaluation seeds (200 episodes each), except ‡, which denotes a single 200-episode evaluation at seed 1000.

Grouping	Train seed	20k	30k
Timestep	1000	78.5‡	79.0‡
Flat chunk	1000	86.5±2.6	77.8±2.0
Flat chunk	1001	86.0±2.5	82.0±2.8
Action dimension	1000	89.0±1.7	90.2±1.6
Action dimension	1001	89.2±2.5	91.2±1.8

Protocol details. Both families use pinned `lerobot/libero` revision `a1aaacb7`, with 1,693 episodes, 273,465 frames, 40 tasks at 10 Hz, and the official evaluator (Liu et al., 2023). Inputs are two 256×256 views, 8-D state, and language; targets are mean/std-normalized 7-D actions. Software is LeRobot 0.5.2 (`8515d456`), MuJoCo 3.3.2, and OSMesa. Batch counts precede Drift's $G = 8$ suffix expansion. SmolVLA uses a pretrained SmolVLM2-500M VLM and a fresh 100M expert; full fine-tuning uses 30k AdamW updates, 10^{-4} learning rate, cosine decay, 1k warm-up, $21 \times 3 = 63$ batch, and bf16 DDP. The $\pi_{0.5}$ pair uses a shared LIBERO-adapted VLM and a fresh 300M expert for 10k updates, $8 \times 7 = 56$ batch, and bf16 DDP; Drift disables language-model gradient checkpointing only during prefix-cache prefill. VLM-frozen grouping uses the same data/schedule, 30k updates, and 21×3 batch, training the fresh expert plus state/action/time projections (about 100M parameters). Within each pair, architecture, initialization, data, update budget, batch, normalization, and cadence are fixed. The $\pi_{0.5}$ pair is not padding-loss-matched: Drift zero-masks padded timesteps; archived FM averages all $H \times D$ entries. Spatial was the development suite; displayed evaluation seeds were not held out. Table 1 uses terminal checkpoints (30k SmolVLA; 10k $\pi_{0.5}$). Execution horizons differ, so comparisons are within family; SDs are recomputed from archived per-seed rates.

Task success and seed replication. On Spatial, three matched training seeds 1000/1001/1002, each evaluated for 200 episodes at fixed evaluation seed 1000, score 96.5/95.0/91.0% for Drift and 94.0/93.5/95.5% for FM: 94.2 ± 2.8 versus $94.3\pm1.0\%$ (sample SD across training seeds). The mean paired difference is -0.2 points and the nominal 95% paired-t confidence interval over the three training-seed pairs is $[-9.6, 9.2]$; we do not claim equivalence or non-inferiority. Table 1 then gives single-training-run breadth checks: Drift's descriptive average point estimate is +2.7 on SmolVLA (Spatial/Object/Goal/Long: +0.3/+8.6/+1.7/+0.3) and +3.2 on the non-padding-loss-matched $\pi_{0.5}$ pair (Long: +6.4); neither establishes objective superiority. Figure 2(a) adds $\pi_{0.5}$ Spatial checkpoints from one training run per objective within each regime: the VLM-frozen gaps are +11.2 at 3k and +5.3 at 6k, while a separate full-fine-tuning pair gives +2.2 at 10k. Frozen/full-FT recipes jointly differ in update budget, trainable modules, and effective conditioning batch (112 versus 56), and frozen FM is still improving at 6k; these are descriptive checkpoint comparisons, not controlled capacity or asymptotic comparisons.

Grouping and implementation. Table 2 reports the only experiment that varies grouping—VLM-frozen SmolVLA on the Spatial development suite. Replication under full fine-tuning and both breadth comparisons test the complete Drift-1 recipe against FM-10, not alternative groupings. In the grouping test, action-dimension grouping has the highest point estimate at 30k in both matched training seeds and stays near 90% at both checkpoints. Timestep grouping is lower but was not replicated across evaluation seeds under the final kernel, so we avoid claiming that it is intrinsically unstable. At training seed 1000, matched flat-chunk runs score 78.3 ± 3.1 at 20k without the fp32 target guard and 86.5 ± 2.6 with it.

Systems and robot feasibility. Figure 2(b) summarizes the timing results. Archived A100 action-generation results report FM-10/Drift-1 = 222.6/50.6 ms for SmolVLA (4.4×; raw output unavailable) and 259.8/85.6 ms for $\pi_{0.5}$ (3.0×). Both use synthetic batch-one inputs and arithmetic means: $\pi_{0.5}$ uses five warm-ups and 30 synchronized calls, whereas the SmolVLA script specifies ten warm-ups and 50 synchronized calls. The $\pi_{0.5}$ microbenchmark switches the FM/Drift inference procedure on one loaded checkpoint; neither pair times every success checkpoint. An archived five-round LeKiwi RTX 2050 summary, whose raw round records are unavailable here, reports Drift-1/FM-10 = $132.4\pm1.2/245.6\pm16.2$ ms (reported SD across round medians; fp32, three 480p views, 50×9 chunks, two warm-ups and eight calls/round). Across the five paired rounds, speedup averages $1.86\times$ (reported SD $0.11\times$). This laptop protocol is not directly comparable to either A100 protocol. For an informal one-task check on the LeKiwi mobile manipulator, models were trained on 60 demonstrations and each policy ran ten salt-bottle pick-and-place trials from the same initial scene: Drift plus KeyStone (K=8 candidates, C=4 clusters; one expert call batched over all candidates) (Dai et al., 2026) scored 9/10, FM-10 7/10, and FM-1 (FM truncated to one Euler step) 3/10. The 132.4-ms timing is Drift without KeyStone; an archived 172-ms KeyStone value lacks raw output. Only Drift used KeyStone, and control cadence and trial logs were not archived. This is feasibility evidence, not a controlled comparison.

5. Limitations and Conclusion

Limitations. $G = 8$ falls outside IDP's single-prediction degeneration result, but our finite-G generated-sample estimator still depends on transient policy samples and is not shown to identify the true conditional drift; we therefore make mechanistic claims only about the tested factorizations within this estimator. Missing same-backbone comparisons include FreqPolicy, FM with a stronger noise shift, and SnapFlow (Su et al., 2025; Chen et al., 2026d; Luan et al., 2026). At 30k, using training seed 1000 and evaluation seed 1000, Drift-1/FM-1/FM-10 Spatial success rates (%) are 89.0/92.5/90.0 (VLM-frozen) and 96.5/91.5/94.0 (full fine-tuning); FM-1 is the one-step truncation of the FM-10 checkpoint. The observed FM-1–FM-10 difference changes sign across these single-seed diagnostics; replication and matched noise-shifted FM remain needed. The $\pi_{0.5}$ results also need a padding-matched FM loss and additional training seeds. Mechanistically, grouping changes Q and S_q as well as channel semantics, so the ablation does not isolate semantic factorization from dimensional concentration or normalization. Action standardization also precludes a simple physical-unit explanation, and factorization could disrupt correlated modes; none of these mechanisms is tested directly. Finally, the latency archive lacks p95, energy, and fully matched end-to-end measurements, and the robot study is small.

Conclusion. On VLM-frozen Spatial, action-dimension grouping stays near 90% at both checkpoints and exceeds flat-chunk point estimates at 30k in both seeds; timestep is not comparably replicated. Three full-fine-tuning seeds give 94.2% for Drift-1 versus 94.3% for FM-10, without equivalence evidence. Four-suite checks are single-run and descriptive; the $\pi_{0.5}$ pair is padding-loss-confounded. One-step inference lowers measured action-generation latency; objective selection remains open.

References

Black, K., Brown, N., Darpinian, J., Dhabalia, K., Driess, D., Esmail, A., Equi, M. R., Finn, C., Fusai, N., Galliker, M. Y., Ghosh, D., Groom, L., Hausman, K., Ichter, B., Jakubczak, S., Jones, T., Ke, L., LeBlanc, D., Levine, S., Li-Bell, A., Mothukuri, M., Nair, S., Pertsch, K., Ren, A. Z., Shi, L. X., Smith, L., Springenberg, J. T., Stachowicz, K., Tanner, J., Vuong, Q., Walke, H., Walling, A., Wang, H., Yu, L., and Zhilinsky, U. $\pi_{0.5}$: a Vision-Language-Action model with open-world generalization. In *Proceedings of the 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pp. 17–40. PMLR, 2025a.

Black, K., Galliker, M. Y., and Levine, S. Real-time execution of action chunking flow policies. In *Advances in Neural Information Processing Systems*, volume 38, pp. 33383–33407. Curran Associates, Inc., 2025b.

Chen, K., Long, Y., Li, S., and Shang, M. ElasticFlow: One-step physics-consistent policy with elastic time horizons for language-guided manipulation. In *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 23685–23701. Association for Computational Linguistics, 2026a. doi: 10.18653/v1/2026.findings-acl.1186.

Chen, Y., Cai, X., Gong, Z., and Huang, Y. High-fidelity one-step generative visuomotor policy via recursive correction, frequency consistency, and contrastive flow matching. *arXiv preprint arXiv:2607.03865*, 2026b.

Chen, Y., Ma, X., and Zhao, B. Mean-Flow based one-step vision-language-action. *arXiv preprint arXiv:2603.01469*, 2026c.

Chen, Y., Zhang, S., Gong, J., and Qiu, X. Let it be simple: One-step action generation for vision-language-action models. *arXiv preprint arXiv:2606.05737*, 2026d.

Chi, C., Feng, S., Du, Y., Xu, Z., Cousineau, E., Burchfiel, B. C. M., and Song, S. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems*, 2023. doi: 10.15607/RSS.2023.XIX.026.

Dai, Y., Chen, Z., Yang, L., and Netravali, R. Geometry guided self-consistency for physical AI. *arXiv preprint arXiv:2605.08638*, 2026.

Deng, M., Li, H., Li, T., Du, Y., and He, K. Generative modeling via drifting. *arXiv preprint arXiv:2602.04770*, 2026.

Gao, Y., Shen, Y., Zhang, S., Yu, W., Duan, Y., Pan, J., Wu, J., Deng, J., and Zhang, Y. Drift-based policy optimization: Native one-step policy learning for online robot control. *arXiv preprint arXiv:2604.03540*, 2026.

Kim, M. J., Finn, C., and Liang, P. Fine-tuning vision-language-action models: Optimizing speed and success. *arXiv preprint arXiv:2502.19645*, 2025.

Koo, J., Park, M., Choi, J., Min, Y., and Sung, M. Drifting field policy: A one-step generative policy via Wasserstein gradient flow. *arXiv preprint arXiv:2605.07727*, 2026.

Li, S., Sun, L., and Chen, Y. One-step flow policy: Self-distillation for fast visuomotor policies. *arXiv preprint arXiv:2603.12480*, 2026.

Liang, H., Chen, X., Wang, B., Chen, M., Liu, Y., Zhang, Y., Chen, Z., Yang, T., Chen, Y., Pang, J., Liu, D., Yang, X., Mu, Y., Shao, W., and Luo, P. MM-ACT: Learn from multimodal parallel generation to act. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 35080–35090, 2026.

Lipman, Y., Chen, R. T. Q., Ben-Hamu, H., Nickel, M., and Le, M. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.

Liu, B., Zhu, Y., Gao, C., Feng, Y., Liu, Q., Zhu, Y., and Stone, P. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. In *Advances in Neural Information Processing Systems*, volume 36, pp. 44776–44791. Curran Associates, Inc., 2023.

Liu, S., Li, B., Ma, K., Wu, L., Tan, H., Ouyang, X., Su, H., and Zhu, J. RDT2: Exploring the scaling limit of UMI data towards zero-shot cross-embodiment generalization. *arXiv preprint arXiv:2602.03310*, 2026a.

Liu, Y., Zhang, S., Dong, Z., Ye, B., Yuan, T., Yu, X., Yin, L., Lu, C., Shi, J., Yu, L. J.-T., Zheng, L., Jiang, T., Gong, J., Qiu, X., and Zhao, H. FASTer: Toward efficient autoregressive vision–language–action modeling via neural action tokenization. In *International Conference on Learning Representations*, 2026b.

Lu, Y., Liu, Z., Fan, X., Yang, Z., Hou, J., Li, J., Ding, K., and Zhao, H. FASTER: Rethinking real-time flow VLAs. *arXiv preprint arXiv:2603.19199*, 2026.

Luan, W., Li, J., Zhao, W., Zhang, W., Wu, T., and Ma, R. SnapFlow: One-step action generation for flow-matching VLAs via progressive self-distillation. *arXiv preprint arXiv:2604.05656*, 2026.

Prasad, A., Lin, K., Wu, J., Zhou, L., and Bohg, J. Consistency policy: Accelerated visuomotor policies via consistency distillation. In *Proceedings of Robotics: Science and Systems*, 2024. doi: 10.15607/RSS.2024.XX.071.

Puthumanaillam, G. and Ornik, M. Amortizing trajectory diffusion with keyed drift fields. *arXiv preprint arXiv:2603.14056*, 2026.

Shukor, M., Aubakirova, D., Capuano, F., Kooijmans, P., Palma, S., Zouitine, A., Aractingi, M., Pascal, C., Russi, M., Marafioti, A., Alibert, S., Cord, M., Wolf, T., and Cadene, R. SmolVLA: A vision-language-action model for affordable and efficient robotics. *arXiv preprint arXiv:2506.01844*, 2025.

Su, Y., Liu, N., Chen, D., Zhao, Z., Wu, K., Li, M., Xu, Z., Che, Z., and Tang, J. FreqPolicy: Efficient flow-based visuomotor policy via frequency consistency. In *Advances in Neural Information Processing Systems*, volume 38, pp. 27769–27797, 2025.

Tarasov, D., Nikulin, A., Zisman, I., Klepach, A., Lyubaykin, N., Polubarov, A., Derevyagin, A., and Kurenkov, V. NinA: Normalizing flows in action. Training VLA models with normalizing flows. *arXiv preprint arXiv:2508.16845*, 2025.

Wang, Z., Li, M., Mandlekar, A., Xu, Z., Fan, J., Narang, Y., Fan, L., Zhu, Y., Balaji, Y., Zhou, M., Liu, M.-Y., and Zeng, Y. One-step diffusion policy: Fast visuomotor policies via diffusion distillation. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 63399–63416. PMLR, 2025.

Xu, C., Zou, Y., Feng, Z., Meng, F., and Liu, S. Ada3Drift: Adaptive training-time drifting for one-step 3D visuomotor robotic manipulation. *arXiv preprint arXiv:2603.11984*, 2026.

Yang, Z., He, Y., Zhong, Y., Zhang, Y., Zhu, X., Mu, Y., Huang, Q., and Ma, Y. Implicit drifting policy: One-step action generation via conditional expert geometry. *arXiv preprint arXiv:2606.01098*, 2026.

Zhan, G., Tao, L., Wang, P., Wang, Y., Li, Y., Chen, Y., Li, H., Tomizuka, M., and Li, S. E. Mean flow policy with instantaneous velocity constraint for one-step action generation. In *International Conference on Learning Representations*, 2026.

Zhang, Y., Ji, K., Zou, Y., Xu, R., Zheng, F., and Cheng, L. Invertible neural network adapter for one-step flow matching in robot manipulation. *arXiv preprint arXiv:2606.19194*, 2026.